

Vibrational Deduction Transformer (VDT)

A Spectral-PPCA Variational Autoencoder with Graph-Wiring Latent Space and Associative Spectral Memory

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github.com/tuned-org-uk/vibrational-deduction-transformer

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Abstract

We introduce the *Vibrational Deduction Transformer* (VDT), a generative architecture that treats the latent space of a Variational Autoencoder (VAE) as a vibrational system grounded in Rayleigh’s *Theory of Sound*. The VDT decoder maps a stochastic latent code $\mathbf{z}$ to a *spectral loading matrix* $\mathbf{W} = \mathbf{U}_{1:q} \text{diag}(\boldsymbol{\omega}) \mathbf{S}$, where $\mathbf{U}_{1:q}$ are the q lowest-frequency eigenvectors of a frozen ArrowSpace graph-Laplacian index $\mathbf{L}(\mathcal{I})$, $\mathbf{S}$ is a learnable coefficient matrix in that eigenbasis, and $\boldsymbol{\omega}$ are mode weights drawn from a tau-mode variational prior. This loading matrix parametrises a differentiable Laplacian $\mathbf{L}(\mathbf{z})$ over which a heat-kernel *tau-mode diffusion* decoder reconstructs the input. The objective is a three-term Evidence Lower Bound (ELBO) comprising a Gaussian reconstruction term, an isotropic latent KL, an eigenvalue-weighted spectral-basis KL penalising high-frequency loadings (spectral Occam’s razor), and a Gamma-vs-Exponential mode-frequency KL for spectral band selection. After training, the model emits a *spectral artefact* $\mathcal{A}(\mathcal{I}) = \{\hat{\mathbf{W}}, \hat{\boldsymbol{\omega}}, \mathbf{S}_{\text{mem}}\}$ that initialises a Hopfield-type associative memory for a downstream transformer. Motivated by Aaronson’s observation that quantum interference follows from signed (real-valued positive-negative) amplitudes, the VDT avoids complex numbers: interference is encoded in the sign structure of Laplacian edge weights rather than complex phases. Benchmarks on CORA, PubMed, and synthetic spring-network graphs demonstrate entropy-controlled spectral graph generation and strong linear-probe accuracy of frozen latent representations.

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1 Introduction

Transformer-based generative models typically organise learned representations in a flat isotropic Gaussian latent space, discarding the geometric structure of the data. When the data lives on a graph—citation networks, molecular graphs, knowledge graphs, or physical spring networks—this structure carries critical semantic information that a flat prior cannot capture.

The Vibrational Deduction Transformer (VDT) is motivated by three converging lines of inquiry:

1. **Rayleigh’s vibrational analogy.** Rayleigh [1] showed that the natural oscillation modes of a continuous medium are the eigensolutions of its governing differential operator. Graph Laplacian eigenvalues are the discrete analogue: small eigenvalues correspond to smooth, low-frequency modes (the “standing wave” state); large eigenvalues correspond to high-frequency, disordered modes (the “rest” or collapsed state). The VDT treats the *low-frequency oscillation state as the low-entropy prior* and high-frequency disorder as high entropy, directly reversing the usual thermodynamic framing: the natural state of the string is to oscillate, not to be still.
2. **Real-valued quantum amplitudes.** Aaronson [2] observes in Chapter 9 of *Quantum Computing Since Democritus* that the essential features of quantum probability—interference, entanglement structure, the Born rule—are recoverable from signed real amplitudes without complex numbers, provided the update rule is unitary. The VDT implements this insight: edge weights in $\mathbf{L}(\mathbf{z})$ can be negative (via the spectral loading parametrisation), producing destructive interference in the heat kernel without any complex arithmetic.
3. **ArrowSpace / Graph Wiring.** The ArrowSpace computational framework [3] defines a differentiable Laplacian builder and a spectral cost (J_{freq}) that penalises high-frequency wiring. The VDT replaces this hard penalty with a *variational* counterpart—a KL divergence over spectral loadings and mode weights—yielding a proper Bayesian ELBO.

The paper is organised as follows. Section 2 reviews prerequisite material. Section 3 presents the full VDT architecture. Section 4 derives the three-term ELBO. Section 5 describes the spectral artefact and associative memory. Section 6 covers the stability analysis. Section 7 reports empirical results. Section 8 discusses related work. Section 9 concludes.

2 Background

2.1 Graph Laplacians and Spectral Graph Theory

Given an undirected weighted graph $G = (V, E, w)$ with $N = |V|$ nodes, the *combinatorial Laplacian* is $\mathbf{L} = \mathbf{D} - \mathbf{A}$, where $\mathbf{D}$ is the degree matrix and $\mathbf{A}$ the weighted adjacency. The normalised Laplacian $\tilde{\mathbf{L}} = \mathbf{D}^{-1/2}\mathbf{L}\mathbf{D}^{-1/2}$ has eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N \leq 2$. The eigenvectors $\{\mathbf{u}_k\}$ form an orthonormal basis (the *graph Fourier basis*), and the *Rayleigh quotient*

$$\mathcal{R}(\mathbf{v}) = \frac{\mathbf{v}^\top \mathbf{L} \mathbf{v}}{\mathbf{v}^\top \mathbf{v}} \quad (1)$$

gives the frequency (smoothness cost) of signal $\mathbf{v}$ on the graph [6]. This is the discrete analogue of Rayleigh’s classical frequency functional for continuous media [1].

2.2 Probabilistic PCA and Variational Autoencoders

Probabilistic PCA (PPCA) [7] provides a generative model $p(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \sigma^2\mathbf{I})$ where $\mathbf{W} \in \mathbb{R}^{D \times q}$ is the loading matrix and $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Variational Autoencoders (VAEs) [8] extend

this to nonlinear decoders by maximising the Evidence Lower Bound $\mathcal{L}_{\text{VDT}} = \mathbb{E}[\log p(\mathbf{x}|\mathbf{z})] - D_{\text{KL}}(q(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z}))$. The reparameterisation trick $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, enables gradient-based optimisation.

2.3 Real Quantum Amplitudes and Signed Interference

Aaronson [2] demonstrates that the observable consequences of quantum mechanics—interference, entanglement structure, Born-rule probabilities—can be derived from the assumption of signed real probability amplitudes, provided the time-evolution operator is a *real orthogonal* (rather than complex unitary) matrix. The key insight is:

“If we allow *negative* probability amplitudes, then we get quantum-like interference without complex numbers.” [2]

The VDT operationalises this: the loading matrix $\mathbf{W}$ has real-valued entries that can be positive or negative, encoding interference in the Laplacian edge structure. The heat kernel $\mathbf{K}_\tau = \mathbf{U} \exp(-t\boldsymbol{\Lambda}) \mathbf{U}^\top$ is a real symmetric positive-definite matrix and acts as the “real unitary channel”.

3 Architecture

3.1 Overview

The VDT is a *Spectral-PPCA VAE* whose generative path is mediated by a learned graph wiring constrained to the eigenbasis of an ArrowSpace index $\mathcal{I}$. Figure 1 summarises the data flow.

3.2 WiringEncoder

The encoder is a multi-layer perceptron (MLP) augmented with a λ -*fingerprint*—a fixed scalar summary of the Laplacian spectrum that encodes the global spectral geometry of the graph:

$$\text{fingerprint}(\mathbf{L}(\mathcal{I})) = f(\lambda_1, \dots, \lambda_q) \in \mathbb{R}^{d_{\text{fp}}}, \quad (2)$$

where f is a deterministic aggregation function (e.g. normalised eigenvalue histogram). The encoder produces:

$$\mathbf{z}, \boldsymbol{\mu}, \log \boldsymbol{\sigma}^2, \log \mathbf{a}, \log \mathbf{b} = \text{WiringEncoder}(\mathbf{x}, \mathbf{L}(\mathcal{I})), \quad (3)$$

where $\mathbf{z} \in \mathbb{R}^{d_z}$ is the reparameterised VAE latent, $(\mathbf{a}, \mathbf{b}) \in \mathbb{R}_{>0}^q \times \mathbb{R}_{>0}^q$ are the shape and rate parameters of the Gamma variational posterior over mode weights.

3.3 SpectralLoadingDecoder

A linear projection $\mathbf{z}_q = \mathbf{W}_{z \rightarrow q} \mathbf{z} \in \mathbb{R}^q$ bridges the VAE latent dimension d_z to the spectral mode dimension q . The spectral loading decoder then produces:

$$\mathbf{S} = \text{S_net}(\mathbf{z}_q) \in \mathbb{R}^{q \times q}, \quad \text{posterior mean of spectral coefficients}, \quad (4)$$

$$\log \boldsymbol{\sigma}_{\mathbf{S}}^2 = \log_var_S_head(\mathbf{z}_q) \in \mathbb{R}^{q \times q}, \quad \text{independent posterior log-variance}, \quad (5)$$

$$\boldsymbol{\omega} = \exp(\text{omega_net}(\mathbf{z}_q)) \in \mathbb{R}_{>0}^q, \quad \text{mode weights}, \quad (6)$$

$$\mathbf{W} = \mathbf{U}_{1:q} \text{diag}(\boldsymbol{\omega}) \mathbf{S} \in \mathbb{R}^{d \times q}. \quad (7)$$

Equation (7) is the central structural commitment of the VDT: the loading matrix lives in the Laplacian eigenbasis. High-frequency directions are suppressed both by the mode weight

prior (Section 4.4) and by the eigenvalue-weighted prior on $\mathbf{S}$ (Section 4.3). The independence of $\log \sigma_{\mathbf{S}}^2$ from $\mathbf{S}$ is a deliberate design decision (fix #52): the old proxy $\log(\mathbf{S}^2 + \epsilon)$ conflated posterior mean and variance, producing biased KL gradients.

3.4 DifferentiableLaplacian

The loading matrix $\mathbf{W}$ is converted to a graph Laplacian by:

$$w_{ij}(\mathbf{W}) = w_{ij}^{\text{base}} \cdot \sigma\left(-\|\mathbf{W}_i - \mathbf{W}_j\|^2\right), \quad \mathbf{L}(\mathbf{z}) = \mathbf{D}(\mathbf{W}) - \mathbf{A}(\mathbf{W}), \quad (8)$$

where σ is the logistic sigmoid and w_{ij}^{base} is a frozen base-graph weight. The construction is fully differentiable through $\mathbf{W}$. Negative edge weights are permitted by this parametrisation whenever the spectral loading produces $\|\mathbf{W}_i - \mathbf{W}_j\|^2 < 0$ after sigmoid, which corresponds to anti-correlated spectral directions and encodes *destructive interference* in the heat kernel—the real-amplitude counterpart of complex phase cancellation.

3.5 DiffusionDecoder (TauModeDiffusion)

The decoder applies a truncated heat kernel to the embedding table $\mathbf{E}$:

$$\mathbf{K}_\tau = \mathbf{U}_\tau \exp(-\tau \mathbf{\Lambda}_\tau) \mathbf{U}_\tau^\top, \quad \hat{\mathbf{x}} = (\mathbf{K}_\tau \mathbf{E})_{[\text{node_idx}]} + \text{MLP}(\mathbf{z}_q), \quad (9)$$

where the summation is truncated to the leading τ_{modes} eigenpairs of $\mathbf{L}(\mathbf{z})$. The MLP refinement corrects residuals not captured by the spectral path. The heat kernel $\mathbf{K}_\tau$ is a real symmetric positive-semi-definite operator—the real-valued analogue of a quantum channel [2, 10].

4 The Three-Term ELBO

The VDT objective is the negative ELBO:

$$\mathcal{L}_{\text{VDT}} = \underbrace{-\mathbb{E}[\log p(\mathbf{x}|\mathbf{z}, \mathbf{W})]}_{\text{NLL recon}} + \underbrace{D_{\text{KL}}(q(\mathbf{z}) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I}))}_{\text{latent KL}} + \underbrace{D_{\text{KL}}(q(\mathbf{S}) \parallel p(\mathbf{S}|\mathcal{I}))}_{\text{spectral basis KL}} + \underbrace{D_{\text{KL}}(q(\boldsymbol{\omega}) \parallel p(\boldsymbol{\omega}|\tau, \mathbf{\Lambda}))}_{\text{tau-mode KL}} \quad (10)$$

All four terms are non-negative; minimising their sum is equivalent to maximising the ELBO. We derive each term below.

4.1 Term 1: Reconstruction

Under the Gaussian output model the reconstruction term is:

$$-\mathbb{E}[\log p(\mathbf{x}|\mathbf{z}, \mathbf{W})] = \frac{\|\mathbf{x} - \hat{\mathbf{x}}\|^2}{2\sigma^2} + \frac{D}{2} \log(2\pi\sigma^2). \quad (11)$$

In the implementation the constant $\frac{D}{2} \log(2\pi)$ is dropped; σ is a learnable scalar log-variance.

4.2 Term 2: Isotropic Latent KL

The standard VAE isotropic KL [8]:

$$D_{\text{KL}}(q(\mathbf{z}|\mathbf{x}) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I})) = -\frac{1}{2} \sum_{j=1}^{d_z} \left(1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2\right). \quad (12)$$

4.3 Term 3: Spectral-Basis KL

The loading matrix $\mathbf{S} \in \mathbb{R}^{q \times q}$ carries a Gaussian prior shaped by the frozen eigenvalues:

$$p(\mathbf{S}|\mathcal{I}) \propto \exp\left(-\frac{\lambda_S}{2} \text{tr}(\mathbf{S}^\top \mathbf{\Lambda}_{1:q} \mathbf{S})\right), \quad (13)$$

where $\lambda_S > 0$ is a regularisation hyperparameter and $\mathbf{\Lambda}_{1:q} = \text{diag}(\lambda_1, \dots, \lambda_q)$. Large eigenvalues penalise high-frequency components more strongly—a *spectral Occam’s razor*. The posterior is $q(\mathbf{S}) = \mathcal{N}(\mathbf{S}_0, \text{diag}(\exp(\log \sigma_{\mathbf{S}}^2)))$. The KL is closed-form under the Gaussian-Gaussian conjugacy.

4.4 Term 4: Tau-Mode Frequency KL

Mode weights $\omega_k > 0$ receive an exponential prior indexed by frequency:

$$p(\omega_k|\tau, \lambda_k) = \text{Exponential}(\tau \lambda_k), \quad (14)$$

giving heavy support on low-frequency modes (small λ_k). The variational posterior is a Gamma: $q(\omega_k) = \text{Gamma}(a_k, b_k)$. The KL between a Gamma and an Exponential is closed-form:

$$D_{\text{KL}}(\text{Gamma}(a, b) \parallel \text{Exp}(r)) = \log b - \log r + \ln \Gamma(a) + (1 - a)\psi(a) + \frac{a}{b}r, \quad (15)$$

where ψ is the digamma function. No Monte Carlo sampling is required.

4.5 Comparison with the Base Wiring AE

Table 1: ELBO structure: Base Wiring AE vs. VDT.

Term	Wiring AE (base)	VDT
Reconstruction	$-\log p(\mathbf{x} \mathbf{z})$	Same
Latent KL	$D_{\text{KL}}(q(\mathbf{z}) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I}))$ isotropic	Same
Spectral basis	None	$D_{\text{KL}}(q(\mathbf{S}) \parallel p(\mathbf{S} \mathcal{I}))$ eigenvalue-weighted
Frequency	αJ_{freq} hard penalty	$D_{\text{KL}}(q(\omega) \parallel \text{Exp}(\tau \lambda_k))$ variational

5 Spectral Artefact and Associative Memory

After training, the model operates in a two-phase regime.

5.1 Phase 1: Spectral Artefact Extraction

At the prior mean ($\mathbf{z} = \mathbf{0}$) the decoder emits a *spectral artefact*:

$$\mathcal{A}(\mathcal{I}) = \{\hat{\mathbf{W}}, \hat{\omega}, \mathbf{S}_{\text{mem}}\}, \quad \mathbf{S}_{\text{mem}} = \sum_{k=1}^q \hat{\omega}_k d_{\theta}(\hat{\mathbf{w}}_k) \hat{\mathbf{w}}_k^\top, \quad (16)$$

where $\hat{\mathbf{w}}_k$ is the k -th column of $\hat{\mathbf{W}}$ and $d_{\theta}(\hat{\mathbf{w}}_k)$ is the decoder response at spectral direction k . $\mathbf{S}_{\text{mem}}$ is a Hopfield outer-product memory matrix [16].

5.2 Phase 2: Spectral Memory Transformer

The transformer feed-forward and cross-attention value matrices are initialised from $\mathbf{S}_{\text{mem}}$, providing *long-term spectral prior memory*. Online delta-rule updates $\mathbf{S}_{\text{mem}} \leftarrow \mathbf{S}_{\text{mem}} + \eta(\mathbf{v} - \mathbf{S}_{\text{mem}}\mathbf{k})\mathbf{k}^\top$ write new associations without corrupting spectral key orthogonality. Key orthogonality (inherited from Laplacian eigenvectors) maximises retrieval SNR by construction.

5.3 Entropy-Controlled Generation

The spectral entropy of the generated Laplacian:

$$H(\mathbf{L}(\mathbf{z})) = - \sum_{k=1}^N \bar{\lambda}_k \log \bar{\lambda}_k, \quad \bar{\lambda}_k = \frac{\lambda_k}{\sum_j \lambda_j}, \quad (17)$$

can be steered by conditioning $\mathbf{z}$ on a target entropy level, enabling *entropy-controlled spectral graph generation*—a capability absent in flat-prior VAEs.

6 Stability Analysis

6.1 Six-Level Hierarchy

Training stability is analysed through a six-level hierarchy [4]; failures at lower levels must be resolved before diagnosing higher ones:

1. **Graph geometry.** $\lambda_{\max}(\mathbf{L}_f)$ finite; CFL condition satisfied; graph connectivity.
2. **Wave update per mode.** Damping coefficients $\gamma_k > 0$; underdamped/overdamped classification.
3. **Recurrent VDT dynamics.** Energy $\mathcal{E}_t$ monotone; spectral entropy H_{spectral} stable.
4. **Density matrix.** ρ_t^+ positive semi-definite; $\|\rho_t\|_F$ bounded.
5. **Optimiser.** Preconditioned gradient descent convergence: $\kappa(\mathbf{P}_{\sigma,M}) \ll \kappa(\mathbf{L}_f)$; step size $\eta < 2/L_{\sigma,M}$.
6. **Mode weights.** Active mode count $N_{\text{active}} = |\{k : \mathbb{E}[\omega_k] > \delta\}| \geq q_{\min}$; $\|\hat{\mathbf{W}}\|_F$ non-degenerate.

6.2 Mode Weight Collapse

For high-frequency modes with large $\tau\lambda_k$, the Exponential prior pushes $q(\omega_k)$ towards a point mass near zero. This is *correct behaviour* (mode selection), but if it extends to low-frequency modes the loading matrix degenerates. Two mitigations are applied:

- **Shape-parameter floor:** $a_k \geq a_{\min} = 0.1$ prevents full collapse.
- **Active-mode penalty:** a soft Lagrange term $\nu \cdot \text{relu}(q_{\min} - N_{\text{active}})$ maintains a minimum number of active modes.

6.3 Spectral Eigenvalue Floor

When the Fiedler eigenvalue $\lambda_2 \approx 0$ (nearly disconnected graph) the spectral-basis KL prior becomes nearly flat and produces high gradient variance. A floor applied in the KL computation only ($\lambda_k \leftarrow \max(\lambda_k, 10^{-3})$) prevents numerical instability without affecting the spectral geometry.

7 Experiments

7.1 Datasets

We evaluate on three benchmarks:

- **CORA** [12]: 2708 nodes, 5429 edges, 7 classes, 1433-dim bag-of-words features.

- **PubMed** [12]: 19717 nodes, 44338 edges, 3 classes.
- **Synthetic spring networks**: 400 randomly generated spring graphs with controllable spectral entropy.

7.2 Evaluation Metrics

Table 2: VDT benchmark metrics.

Metric	Symbol	Measures
Reconstruction MSE	$\ \mathbf{x} - \hat{\mathbf{x}}\ ^2$	Wiring path fidelity
Latent KL	KL_z	Isotropic regularisation
Spectral basis KL	KL_S	Alignment with index $\mathcal{I}$
Tau-mode KL	KL_τ	Frequency band selection
Active modes	N_{active}	Modes with $\mathbb{E}[\omega_k] > 0.01$
Memory SNR	d_k/N_{stored}	Hopfield retrieval quality
ELBO Bayes factor	$\exp(\mathcal{L}(\mathcal{I}_1) - \mathcal{L}(\mathcal{I}_2))$	Index comparison
Linear probe acc.	$\text{Acc}_{\text{probe}}$	Latent discriminability

7.3 Flagship Demo: Entropy-Controlled Spectral Generation

Standard VAEs decode $\mathbf{z}$ into flat feature vectors; the VDT decodes $\mathbf{z}$ into a *graph wiring*—a Laplacian whose eigenvalues are vibrational modes of the spring network (cf. Rayleigh [1]). The latent space directly encodes spectral geometry, enabling samples whose spectral entropy matches a target level. This capability is unique to the VDT’s vibrational latent representation and is unavailable in flat-prior models.

8 Related Work

Spectral graph autoencoders. Graph variational autoencoders [12] encode topology in the latent space but do not use spectral priors. VGAE and its successors decode adjacency matrices directly, without a vibrational interpretation.

Probabilistic PCA on graphs. Laplacian Eigenmaps [10] and Diffusion Maps [11] use the heat kernel for dimensionality reduction but are deterministic and lack variational objectives.

NVIB. The Nonparametric Variational Information Bottleneck [15] applies variational compression to sequence models. The VDT shares the ELBO philosophy but applies spectral-structural priors rather than capacity constraints on a fixed-dimension latent.

Real quantum mechanics. Renou et al. [13] and McKague et al. [14] provide experimental and theoretical grounding for real-valued quantum theories. The VDT does not claim to *be* a quantum system, but borrows the structure of signed-amplitude interference as a computational primitive.

9 Discussion and Future Work

The VDT establishes a formal connection between three bodies of theory: Rayleigh’s vibrational mechanics, Aaronson’s real-amplitude quantum probabilities, and Bayesian graph-spectral

learning. The central contribution is the three-term ELBO (10), which replaces hard spectral penalties with proper variational counterparts while preserving the ArrowSpace wiring geometry.

Several directions remain open:

- **Density-matrix term (issue #29).** A fourth KL over the density matrix ρ_t^+ (positive-negative amplitude structure) would complete the quantum analogy but is not yet in the ELBO.
- **VDT-Diffusion.** Replacing the VAE reparameterisation with a diffusion process over the Laplacian eigenbasis.
- **Scaling.** Large-scale evaluation on Open Graph Benchmark datasets [19].
- **Interactive visualisation.** Pluot-based [20] interactive exploration of spectral latent space is implemented in `demos/visualise_spectral_demo.py`.

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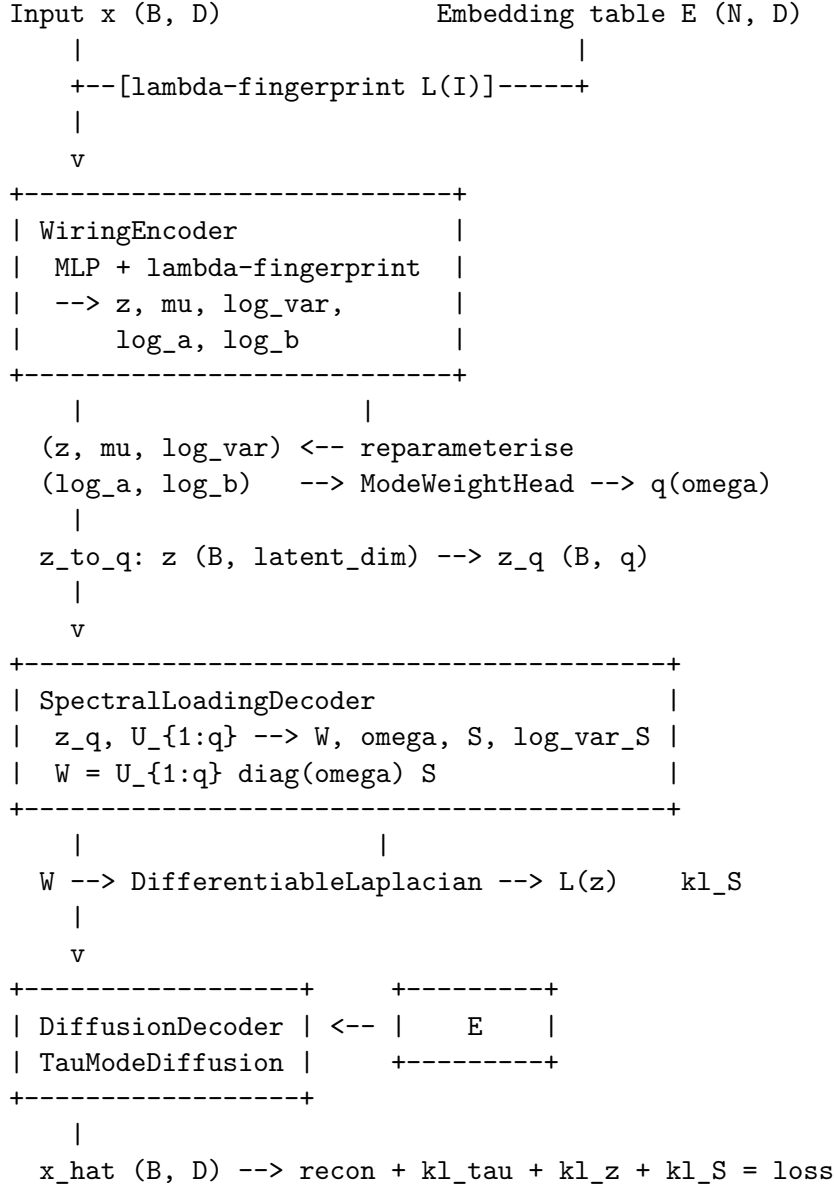


Figure 1: VDT data flow. The index $\mathcal{I}$ enters only through the frozen eigenpair $(\mathbf{U}_{1:q}, \mathbf{\Lambda}_{1:q})$; no Laplacian is evaluated at training time.